Conceptual article (research framework) · 18 October 2026

Artificial intelligence in SOC alert triage

The moderating role of cyber hygiene, Zero Trust and tool fragmentation

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Abstract

The effect of artificial intelligence (AI) on alert triage in Security Operations Centers (SOCs) depends on the quality of the environment in which it is deployed. This article proposes a conceptual framework to examine that thesis. Its research question is: whether and how the maturity of cyber hygiene, the consistency of Zero Trust implementation and the degree of tool fragmentation modify the effect of AI, relative to conventional triage, on detection quality and on investigation cost.

The framework distinguishes information volume from information noise. It treats the use of AI as the central variable, the three factors as moderators, and alert volume and rule-configuration complexity as control variables. Four research propositions are formulated, each with its expected direction and its refutation criterion. A two-phase validation protocol is proposed: an exploratory qualitative study, followed by a multilevel analysis complemented by a replay of labelled alerts.

The article reports no empirical results. It aims to provide a testable basis for determining the conditions under which AI genuinely reduces information noise rather than automating existing dysfunctions.

Keywords: SOC, alert triage, information noise, artificial intelligence, cyber hygiene, Zero Trust, tool fragmentation, moderation effect, human decision-making.

1. Introduction

SOCs concentrate the monitoring, detection, analysis and response to incidents. They rely on solutions for security event management, endpoint detection, threat intelligence and response orchestration. The multiplication of cloud environments, the spread of digital identities and the growth of telemetry sources increase the volume of data to be exploited, and organisations stack specialised tools to cover ever wider perimeters.

The literature describes the human consequences of this situation. Tariq et al. (2025) survey the research on alert fatigue in SOCs. Alahmadi et al. (2022) and Kokulu et al. (2019) report, from interviews with analysts, the difficulties linked to the relevance of alarms and to the fit between tools, processes and people. Vielberth et al. (2020) offer a systematic study of SOCs and their open challenges. Ofte and Katsikas (2023) examine operators' situation awareness.

Faced with these difficulties, AI is increasingly presented as a lever for prioritising, correlating and summarising alerts. Chhetri et al. (2024) study cooperation between analysts and AI in this respect. But AI is inserted into an already structured environment, with its detection rules, its tools and its security practices, and nothing allows us to assume that its contribution is the same from one SOC to another.

1.1 The gap addressed

The works cited most often examine alert load, human-AI cooperation or situation awareness. The link between the effect of AI and the quality of the technical and organisational environment is less formalised. The working hypothesis of this article is that this gap exists and is important; the targeted review described in section 5 must confirm or refute this before any submission.

A triage AI consumes context: asset identifiers, owners, account history, patch status, endpoint posture, logs from several tools. If this context is incomplete, inconsistent or scattered, the AI may go faster without deciding better. Conversely, a well-controlled environment may allow the AI to produce a real gain. Context is therefore not a mere confounding variable: it may condition the very effect we seek to measure.

1.2 Research question

How do the maturity of cyber hygiene, the consistency of Zero Trust implementation and the degree of tool fragmentation modify the effect of artificial intelligence, relative to conventional triage, on detection quality and on investigation cost in SOC

alert triage?

Three sub-questions break it down:

1. **SQ1 (hygiene).** Is the advantage of AI over conventional triage greater when cyber hygiene is more mature?
2. **SQ2 (fragmentation).** Does this advantage diminish when tools are fragmented and poorly interoperable, or does AI on the contrary compensate for this fragmentation?
3. **SQ3 (Zero Trust).** For alerts related to identities and access, does a consistent Zero Trust implementation increase the advantage of AI?

Both outcomes are examined together: a time saving obtained at the price of missed threats is not an improvement.

1.3 Contributions and outline

The article provides three elements: a functional definition of information noise distinct from alert volume and alert fatigue; a moderation framework in which AI is the central variable and three characteristics of the environment are moderators; and four falsifiable propositions with a validation protocol. Section 2 reviews the literature, section 3 presents the framework, section 4 formulates the propositions, section 5 describes the validation method, and sections 6 to 8 cover the discussion, the limitations and the conclusion.

2. Literature review

This review is deliberately targeted at what underpins the model. It remains preliminary: section 5 provides for a reproducible review before submission.

2.1 Alert volume and fatigue

Alert fatigue refers to the degradation of attention or vigilance that may appear when analysts handle many alerts whose relevance is low or hard to establish (Tariq et al., 2025). The number of alerts measures load, not the difficulty of the work: two SOCs receiving the same number of alerts may face very different efforts depending on available context, redundancy and investigation procedures. A distinction must therefore be made between the volume of events collected, the volume of alerts generated, the proportion of relevant alerts, the difficulty of qualification and the quality of final decisions.

2.2 AI in triage and human-AI cooperation

AI can prioritise alerts, link events, summarise context and automate repetitive tasks. Chhetri et al. (2024) argue for cooperation in which automation handles suitable tasks and human expertise remains engaged for ambiguous cases. Work on automation shows, however, that overreliance and complacency are known risks (Parasuraman and Riley, 1997; Parasuraman and Manzey, 2010). An AI that processes more alerts in less time therefore improves security only if the risk of missing a relevant threat does not increase. This caveat justifies measuring investigation cost and detection quality together.

2.3 Tool fragmentation and loss of context

SOCs combine several specialised technologies. This diversity improves coverage but may degrade integration: different identifiers for the same asset, redundant alerts, context to be reconstructed manually across several interfaces. Kokulu et al. (2019) document mismatches between technologies, processes and people in SOCs. The relevant variable is not the number of products but interoperability, the semantic consistency of data and the sharing of context between components. An environment made up of many well-integrated tools is not fragmented in the sense of this model.

2.4 Cyber hygiene

Cyber hygiene covers the basic practices that keep an environment under control: asset inventory, patch management, administration of accounts and privileges, secure configuration. ANSSI (2017) offers a reference set of measures. For a SOC, these practices determine how easily an alert can be tied to an asset and to an owner, and the stability of baseline behaviours. An improvement in hygiene may, however, reveal previously invisible events and temporarily increase detections; the expected effect therefore concerns the interpretability of alerts, not their number.

2.5 Zero Trust

Zero Trust rests on explicit verification of access, least privilege, and the assessment of context and risk at each request (Rose et al., 2020). The CISA (2023) maturity model structures its implementation in pillars, including identity and devices. The associated controls produce information on identities, devices and access conditions which, correlated with other sources, can help qualify an access alert. But poorly harmonised policies can also multiply denials and context switches. Zero Trust is therefore neither an intrinsic source of noise nor a guarantee of noise reduction. Its role in this model is restricted to alerts related to identities and access, where it provides context.

2.6 Summary and current support

The following table indicates, for each element of the model, the assumed role and the degree of documentary support currently available to this article. It highlights the parts to be consolidated by the targeted review.

Element	Assumed role in the model	Current documentary support
Alert fatigue and volume	Control variable	Strong (Tariq et al., 2025; Alahmadi et al., 2022)
Human-AI cooperation	Framework of assisted triage	Partial (Chhetri et al., 2024; Parasuraman and Manzey, 2010)
Tool fragmentation	Moderator	Partial, mostly qualitative (Kokulu et al., 2019)
Cyber hygiene	Moderator	Reference frameworks (ANSSI, 2017), little work on the link with the SOC
Zero Trust	Moderator, identity and access alerts	Reference frameworks (Rose et al., 2020; CISA, 2023), little work on the link with the SOC
Interaction between these factors and AI	Core of the contribution	To be established by the targeted review

The gap is therefore less the absence of literature on each factor than the absence, to our knowledge, of a framework that articulates them as moderators of the effect of AI on triage.

3. Conceptual framework

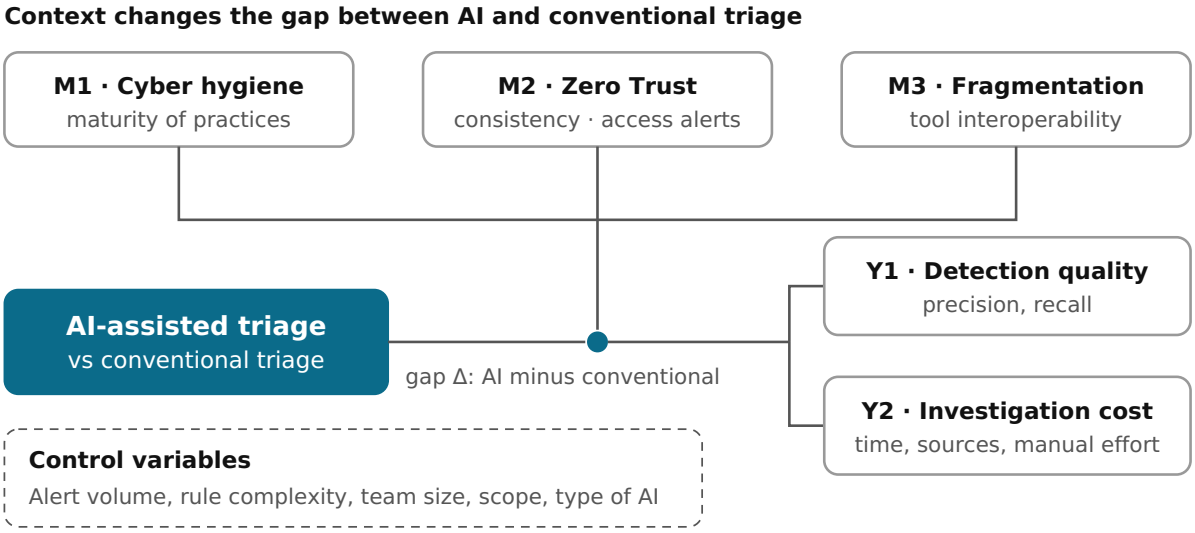
3.1 Definition of information noise

Information noise refers to the share of security information that, in a given operational context, mobilises analysis resources without contributing proportionally to the identification, qualification or handling of relevant threats.

This definition is functional. A non-malicious alert is not necessarily noise: it may verify the coverage of a control or reveal a misconfiguration. Noise is judged in light of the purpose of the investigation, the value of the signal and the effort needed to exploit it. It differs from volume, which is a quantity, and from alert fatigue, which is a possible human consequence of repeated exposure to a load that is hard to manage.

3.2 Model variables

The model poses a question about a gap: does AI do better or more cheaply than conventional triage, and does this gap vary with the environment? It retains one central variable, three moderators, two outcomes and a set of control variables. The figure below gives its structure and the table the working definitions.



moderation model · 1 central variable, 3 moderators, 2 outcomes

The three moderators act on the gap between AI and conventional triage, not directly on the outcomes; the control variables act on the two conditions compared.

Role	Variable	Working definition
Central	AI-assisted triage	Use of AI to prioritise, correlate or summarise alerts, compared with conventional triage based on rules and manual procedures. Described by its functions, its level of automation and human supervision.
Moderator M1	Cyber hygiene maturity	Completeness of the asset inventory, patch compliance, control of accounts and privileges, maintained configurations.
Moderator M2	Consistency of Zero Trust implementation	Harmonisation of access policies and integration of identity and device data. Applied to alerts related to identities and access.
Moderator M3	Tool fragmentation	Low effective interoperability: no common identifiers for the same asset, scattered context, number of interfaces to consult per alert.
Outcome Y1	Detection quality	Precision, recall and false-negative rate on relevant alerts.
Outcome Y2	Investigation cost	Qualification time, number of sources consulted, manual effort per alert.
Control	Volume and configuration	Alert volume, rule-configuration complexity, team size and experience, monitored scope, type of AI.

3.3 Reading the model

Let Δ be the performance gap between AI-assisted triage and conventional triage, measured separately on Y1 and on Y2. A moderator is a variable that makes Δ vary: the effect of AI is not assumed constant, it is assumed conditional. The model does not say that AI is effective or ineffective in general; it says where and under what conditions its effect changes.

Three clarifications frame this reading.

1. Maturity and consistency are two distinct constructs. Hygiene is measured by a maturity of practices, Zero Trust by the consistency of an architecture and its policies. They will be operationalised separately (section 5).
2. Volume and rule complexity are controls, not objects of study. They influence the alert load in both conditions compared, but the framework does not seek to explain their own effect.
3. Competing readings exist. Hygiene may also act as a factor that changes the number of alerts (direct effect), and the most mature SOCs may be those that adopt AI first (selection bias). The protocol in section 5 provides for within-SOC comparisons to limit this second risk.

4. Research propositions

The four propositions below are theoretical statements to be tested, not results. Each specifies the expected effect, the mechanism and what would refute it. The “advantage of AI” refers, for each outcome, to the gap in favour of AI relative to conventional triage: higher detection quality or lower investigation cost.

P1. Baseline effect: the gain is first about cost

Statement. On average, AI-assisted triage reduces investigation cost (Y2) relative to conventional triage, but this reduction does not suffice to establish an improvement in detection quality (Y1), whose gap may be null, positive or negative.

Mechanism. AI acts first on the speed of qualification through prioritisation and summarisation. Detection quality depends on the accuracy of rankings and on the risk of masking an important alert, a known risk of decision-support systems (Parasuraman and Manzey, 2010).

Refutation. P1 is refuted if cost is not lower under AI, or if the advantage on Y1 is uniformly positive in all environments, which would make it pointless to distinguish the two outcomes.

P2. Cyber hygiene: the advantage grows with maturity

Statement. With alert volume controlled, the advantage of AI on Y1 and on Y2 is larger the higher the maturity of cyber hygiene (M1).

Mechanism. A complete inventory, controlled accounts and stable configurations allow the AI to tie each alert to an asset and an owner, and to rely on stable baseline behaviours.

Caveat. Better hygiene may reveal previously invisible events and temporarily increase the number of detections; hence the control for volume.

Refutation. The interaction between the use of AI and M1 is null or of opposite sign after controlling for volume.

P3. Tool fragmentation: the advantage shrinks, or AI compensates

Statement. The advantage of AI on Y1 and on Y2 is smaller the more tools are fragmented and poorly interoperable (M3).

Mechanism. Inconsistent identifiers and scattered context limit the AI’s ability to link data; its outputs then rest on partial context.

Competing proposition. AI could on the contrary compensate for fragmentation by correlating scattered sources that the analyst would otherwise have to consult one by one. The advantage would then be larger in a fragmented environment. These two readings oppose each other and can be told apart by the data; deciding between them is a possible contribution of the study.

Refutation. The interaction between the use of AI and M3 is null or positive. A positive interaction would support the competing reading.

P4. Zero Trust: the advantage grows with consistency, for access alerts

Statement. For alerts related to identities and access, the advantage of AI on Y1 and on Y2 is larger the more consistent the Zero Trust implementation (M2) is, that is, the more policies are harmonised and identity and device data are integrated.

Mechanism. Zero Trust controls produce usable context for qualifying an access alert: device posture, session risk, the access decision applied.

Caveat. Poorly harmonised policies may multiply denials and context switches, and reduce the contribution of Zero Trust to triage.

Refutation. No interaction for these alerts, or an interaction of the same size for alerts unrelated to access. This second case would indicate that the measured construct reflects general maturity rather than Zero Trust.

Summary

The combination of the three moderators (higher-order interactions) is not formulated as a proposition: it is treated as an exploratory question in section 7.

Proposition	Factor	Expected effect on the advantage of AI	Alerts concerned

P1	None (baseline effect)	Advantage on Y2; advantage on Y1 not established	All
P2	M1 Cyber hygiene	Grows with maturity	All
P3	M3 Tool fragmentation	Shrinks with fragmentation (competing reading: grows)	All
P4	M2 Zero Trust	Grows with consistency	Identity and access

5. Proposed validation methodology

Validation follows a three-phase sequence, from the most exploratory to the most constraining. The same logic runs through it: comparing AI and conventional triage within contexts that differ in their hygiene, their Zero Trust and their fragmentation.

5.1 Phase 0. Reproducible targeted review

A restricted systematic review, reported according to PRISMA 2020 (Page et al., 2021), will verify the gap announced in section 1.1. It will query the intersections between SOC, alert triage, AI, cyber hygiene, Zero Trust and tool integration, and will document the databases, search strings, inclusion criteria and selection. Its result conditions the final formulation of the contribution.

5.2 Phase 1. Exploratory qualitative study

Semi-structured interviews will be conducted with SOC analysts, detection leads, CISOs and tool-integration specialists. A target of about 12 to 20 interviews in 4 to 6 different SOCs (in-house, service providers, public and private sector) is proposed, until thematic saturation. The interviews will check that the moderators correspond to real mechanisms and that the working definitions are operational, and will collect examples of cases where AI helped or hindered triage.

5.3 Phase 2. Multilevel analysis and controlled replay

Multilevel analysis. Alerts (level 1) are nested within SOCs or client perimeters (level 2). The moderators M1, M2 and M3 are measured at level 2; the use of AI varies within a single SOC where possible (before and after deployment, processing queues with and without AI). Mixed models are estimated with interaction terms between level 1 and level 2 (Raudenbush and Bryk, 2002; Hayes, 2018). The power of level-2 interactions depends on the number of SOCs or perimeters, not on the number of alerts. If this number is insufficient to estimate three interactions, the study will retain two moderators, and managed service providers, which supervise several clients at different maturity levels, are a source of natural variation to explore.

Controlled replay. A set of alerts labelled by experts serves as ground truth. Analysts process it with or without AI assistance, and the quality of the context provided is varied (complete or degraded inventory, common identifiers or not, identity data integrated or not). This manipulation makes it possible to approach a causal effect of the moderators on the advantage of AI, which production data alone cannot establish. Injected scenarios (for example from attack-simulation exercises) complete the design to estimate recall, which is not observable in production.

5.4 Operationalisation

Indicators will be fixed before data collection. A drop in the number of alerts or in processing time will be interpreted as a gain only if recall is not degraded.

Construct	Envisaged indicators	Source
AI-assisted triage	Assisted functions, level of automation, human supervision, share of recommendations corrected	Interviews, tool logs
M1 Hygiene	Composite score inspired by the ANSSI (2017) measures: inventory coverage, patch compliance, review of accounts and privileges	Audit, asset-management data
M2 Zero Trust	Scale inspired by the identity and device pillars of the CISA (2023) model and the principles of Rose et al. (2020): policy consistency, data integration	Audit, architecture documentation
M3 Fragmentation	Number of interfaces consulted per alert, share of assets resolved under a common identifier, time to obtain context	Observation, logs
Y1 Detection quality	Precision, recall, false-negative rate	Labelled set, injected scenarios

Y2 Investigation cost	Median qualification time, number of cross-consultations, manual effort	Logs, timing
Controls	Alert volume, rule complexity, team size and experience, scope, type of AI	Operational data

5.5 Analysis strategy

A descriptive analysis will characterise volumes, distributions and differences between contexts. Mixed models will test the interaction between the use of AI and each moderator, with random effects by SOC and control variables; P4 will be estimated on identity and access alerts only, then compared with other alerts as a specificity test. Sensitivity analyses will vary the definition of the indicators, and multiple comparisons will be corrected. Since AI deployment is not random and the most mature SOCs may adopt it first, maturity is modelled explicitly and within-SOC comparisons are preferred. No result can be anticipated before data collection.

5.6 Ethics and confidentiality

Alert data are sensitive and may contain personal data. The study will require confidentiality agreements with the organisations, anonymisation or pseudonymisation of logs, and aggregated data for processing times so as not to evaluate analysts individually. Interviews will be subject to informed consent.

6. Discussion

6.1 Theoretical contributions

The framework shifts the research question from volume to the conditional value of AI. Instead of asking whether AI reduces alert fatigue, it asks in which context AI improves detection and reduces investigation cost. This reformulation has three consequences.

1. AI is treated as a processing capability inserted into an environment, not as an isolated solution. Its effect is conditional, and it is this conditionality that is tested.
2. A productivity gain is not a security gain. By measuring Y1 and Y2 together, the framework prevents concluding that an AI performs well when it reduces processing time while letting more threats through.
3. A competing reading is formulated from the outset. For fragmentation, P3 and its competing proposition predict opposite signs, which makes the framework falsifiable on a point where practitioners' intuitions diverge.

6.2 Implications for cyber governance

Security and SOC managers can draw from the framework a diagnostic grid before investing in a layer of AI. Do the observed difficulties stem from poorly calibrated rules, an incomplete inventory, poorly controlled accounts, non-interoperable tools or a lack of procedures? If the answer lies in the environment, AI risks producing a gain in speed without a gain in detection, and making it harder to spot the defects it automates.

The framework also suggests complementing volume indicators with measures of relevance, context, qualification cost and decision performance, and reserving ambiguous cases for human supervision. It does not oppose innovation: it seeks to determine the conditions under which innovation produces a measurable improvement.

6.3 Sequencing of investments

If P2 and P4 hold, cyber hygiene and the consistency of Zero Trust are not only autonomous security objectives: they increase the return on investments in detection AI. If P3 is supported, tool integration comes before AI. If the competing proposition prevails, AI may on the contrary be a short-term palliative for fragmentation. In both cases, the investment decision benefits from relying on a diagnosis of the context and not on the tool's promise alone.

7. Limitations and perspectives

7.1 Limitations

Status of the manuscript. The article is conceptual. It rests on no original data and its propositions are not established. The literature review is preliminary and must be replaced by the reproducible review of phase 0.

Construct validity. Hygiene, Zero Trust consistency and fragmentation are composite constructs whose measurement depends on context. The proposed scales are inspired by frameworks (ANSSI, 2017; CISA, 2023) that were not designed to measure moderators of AI effectiveness and will need to be validated.

Scope of Zero Trust. The moderator M2 applies only to identity and access alerts. It says nothing about the triage of other families of alerts.

Ground truth. Precision and recall require a reliable reference on relevant events. In production, recall is not observable and relies on injected scenarios, whose representativeness is limited.

Selection and causality. AI deployment is not random. Within-SOC comparisons and the controlled replay limit the bias without eliminating it.

Heterogeneity of AI. Machine-learning classifiers, correlation engines and assistants based on language models have neither the same strengths nor the same weaknesses, and evolve quickly. The type of AI is a control, but its results may not transfer from one generation of tools to the next.

Generalisation. In-house SOCs, service providers and public bodies differ in their resources and objectives. Comparisons must take this into account.

7.2 Perspectives

Higher-order interactions. Do the effects of the three moderators add up, offset each other, or is one of them enough? The question is exploratory and requires a larger sample than that of a first study.

Temporal dimension. Maturity evolves. A longitudinal study could track the advantage of AI as hygiene, Zero Trust or tool integration improve.

Human factors. The effect of human supervision, overreliance and complacency on Y1 could be modelled as an additional mediator or moderator.

Adversary adaptation. A triage AI may become a target. Extending the framework to robustness against attacks aimed at detection deserves work of its own.

8. Conclusion

This article proposes shifting the analysis of alert fatigue from volume to the conditional value of artificial intelligence. It formulates a precise question: whether and how cyber hygiene, the consistency of Zero Trust and tool fragmentation modify the effect of AI on triage, in terms of detection quality and investigation cost.

The central proposition is that the operational value of AI in a SOC depends not only on its own capabilities but also on the quality of configuration, integration and contextualisation of the environment in which it operates. Four falsifiable propositions, one of which opposes two readings of the effect of fragmentation, and a three-phase protocol make it possible to put it to the test.

If it holds, the question to ask before adding a layer of AI is no longer only what the tool can do, but what the environment allows it to do. If it is refuted, the framework will at least have clarified the conditions under which AI brings a measurable gain independently of context, which would also be a useful result.

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